

Midterm 3

● Graded

Student

Victoria Worotniak

Total Points

78 / 100 pts

Question 1

Question 1

26 / 30 pts

1.1 a

6 / 10 pts

– 0 pts Correct

– 2 pts first is wrong

✓ – 2 pts second is wrong

✓ – 2 pts third is wrong

– 2 pts fourth is wrong

– 2 pts fifth is wrong

1.2 b

10 / 10 pts

✓ – 0 pts Correct

– 2 pts h_1 should be 3093.9 kJ/kg

– 0 pts $h_1 = h_2$

– 10 pts not done or wrong

– 2 pts interpolation is wrong

– 2 pts T_1 is not equal to T_2

1.3 c

10 / 10 pts

✓ – 0 pts Correct

– 6 pts equation is wrong

– 1 pt calculation error

– 10 pts not done

– 4 pts ΔT are wrong

Question 2

Question 2

24 / 30 pts

2.1 — a

4 / 4 pts

✓ - 0 pts Correct

- 2 pts not shell and tube

- 3 pts inlets and outlets are missing or wrong

- 4 pts not done

2.2 — b

5 / 5 pts

✓ - 0 pts Correct

- 1 pt control volume

- 5 pts not done

2.3 — c

6 / 6 pts

✓ - 0 pts Correct

- 3 pts mass balance missing or wrong (should be $m_1(\dot{)} + m_2(\dot{)} = m_3(\dot{)}$)

- 2 pts simplified energy balance is wrong (should be $0 = m_1(\dot{)}h_1 + m_2(\dot{)}h_2 - m_3(\dot{)}h_3$)

- 1 pt error in energy balance

- 6 pts not done

2.4 — d

5 / 10 pts

- 0 pts Correct

- 1 pt $h_1 = 3188.4$

- 1 pt $h_2 = 251.13$

- 1 pt $h_3 = 2748.7$

✓ - 5 pts equation is wrong

- 1 pt $m_1(\dot{)} = 333.3 \text{ kg/s}$

- 10 pts not done or wrong

- 1 pt calculation error

2.5  e

4 / 5 pts

– 0 pts Correct

✓ – 1 pt calculation error

– 3 pts Equation is wrong ($m_1(\dot{})+m_2(\dot{})=m_3(\dot{})$)

– 5 pts not done

Question 3

Question 3

28 / 40 pts

3.1 a

5 / 5 pts

✓ - 0 pts Correct

- 5 pts not done

3.2 b

3 / 3 pts

✓ - 0 pts Correct

- 1 pt closed system

- 1 pt ideal gas

- 1 pt not control volume

- 3 pts not done

3.3 c

5 / 5 pts

✓ - 0 pts Correct

- 3 pts closed system energy balance

- 5 pts not done

3.4 d

8 / 10 pts

- 0 pts Correct

✓ - 2 pts COP is wrong ($COP = T_h / (T_h - T_c) = 2$)

- 2 pts $Q_h = 250 \cdot 1.5 = 375$

- 2 pts $COP = Q_h / W \quad 2 = 375 / W$ thus $W = 187.5$

- 10 pts not done

3.5 e

0 / 10 pts

- 0 pts Correct

✓ - 10 pts not done or wrong

- 2 pts $T = 300K$

- 1 pt $P_2 = 76.1 kPa$

- 7 pts equation is right, but not done

✓ - 0 pts Correct

- 7 pts not done or wrong

- 2 pts adiabatic parts, specific volume is not the same

- 2 pts isothermal parts, P is not the same

NAME: Victoria WorotniakPlease fill in answers in the space supplied and show all your work to receive full credit.1. a) (10 pts) State whether the following statements are **True or False**

T A diffuser is a flow passage of varying cross-sectional area in which the pressure of a gas or liquid increases in the direction of flow.



F Enthalpy for the inlet of the turbine is greater than the outlet of the turbine.

T A pump is an equipment to increase the pressure or to elevate gas.

F Turbine is an example of steady flow device that result in an increase in working fluid pressure from inlet to exit.

T The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

b) (10 pts) Steam at 1MPa and 320°C is throttled adiabatically to a pressure of 0.7 bar. If the change in kinetic energy is negligible, calculate the temperature and specific volume of the steam after throttling.



$$\dot{m}_i h_i = \dot{m}_e h_e$$

$$h_i = h_e$$

$$\dot{m}_i = \dot{m}_e$$

$$1 \text{ MPa} = 10 \text{ bar}$$

<u>T</u>	<u>h</u>
280	3035
2	3093.9
320	3115.3

$$h_i @ 10 \text{ bar}, 320^\circ\text{C} = 3093.9 \text{ kJ/kg}$$

$$h_e = 3093.9 \text{ kJ/kg} @ 0.7 \text{ bar}$$

$$T = 280 + \frac{3093.9 - 3035}{3115.3 - 3035} (320 - 280) = \boxed{309^\circ\text{C}}$$

<u>v</u>	<u>h</u>
3.640	3035
2	3093.9
3.905	3115.3

$$v = 3.640 + \frac{3093.9 - 3035}{3115.3 - 3035} (3.905 - 3.640) = \boxed{3.83 \text{ m}^3/\text{kg}}$$

c) (10 points) In a shower, cold water at 10°C flowing at a rate of 7kg/min is mixed with hot water at 70°C flowing at a rate of 3kg/min. Calculate the exit temperature of the mixture.

$$Q = \dot{m} c_p \Delta T$$

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

$$3 + 7 = 10$$

$$\dot{m}_i h_i = \dot{m}_e h_e$$

$$\dot{m}_1 c_p \Delta T = \dot{m}_2 c_p \Delta T$$

$$\dot{m}_1 (T_2 - T_1) + \dot{m}_2 (T_2 - T_1) = \dot{m}_3 (T_2 - T_1)$$

$$\dot{m}_1 T_1 + \dot{m}_2 T_2 = \dot{m}_3 T_F$$

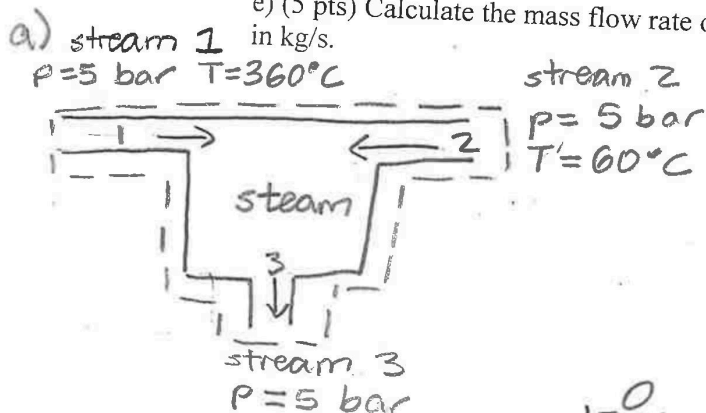
$$(7 \text{ kg/min})(10^\circ\text{C}) + (3 \text{ kg/min})(70^\circ\text{C}) = (10 \text{ kg/min}) T_F$$

$$\boxed{T_F = 28^\circ\text{C}}$$

2. (30 pts)

A stream of superheated steam at 5 bar and 360 °C with a mass flow rate of 20,000 kg/min (stream 1) is mixed with a stream of liquid water at 5 bar and 60 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 5 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy.

- (4 pts) Draw a schematic of the system and label the system boundary.
- (5 pts) State the engineering model.
- (6 pts) Write the simplified balances for mass and energy.
- (10 pts) Calculate the mass flow rate of the liquid water entering the mixing chamber, in kg/s.
- (5 pts) Calculate the mass flow rate of the saturated vapor exiting the mixing chamber, in kg/s.



b) Engineering model:

- Steady state • $\dot{Q} = 0$
- Control volume
- $\Delta KE = \Delta PE = 0$
- 2 inlet, 1 outlet

c) $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$

adiabatic $\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i (h_i + \frac{V_i^2}{2} + g z_i) - \sum \dot{m}_e (h_e + \frac{V_e^2}{2} + g z_e)$

$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$

$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e$

$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$

$h_3 @ 5 \text{ bar} = 2748.7 \text{ kJ/kg}$
(sat. vapor)

d) $h_1 @ 5 \text{ bar}, 360^\circ\text{C} = 3188.4 \text{ kJ/kg}$

$h_2 @ 5 \text{ bar}, 60^\circ\text{C} = 251.13 \text{ kJ/kg}$

$\dot{m}_1 = 20,000 \text{ kg/min} = 333.33 \text{ kg/s}$

$333.33 \text{ kg/s} (3188.4 \text{ kJ/kg}) = \dot{m}_2 (251.13 \text{ kJ/kg})$

$\dot{m}_2 = \boxed{4232.07 \text{ kg/s}}$

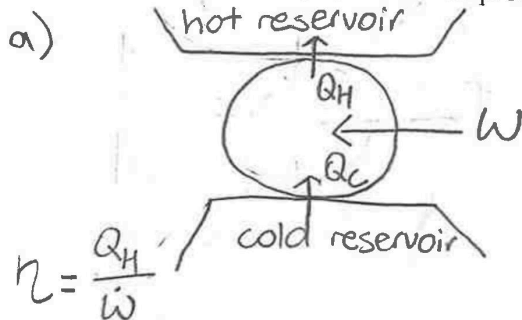
e) $\dot{m}_1 + \dot{m}_2 = \dot{m}_3$

$333.33 \text{ kg/s} + 4232.07 \text{ kg/s} = \boxed{4565.40 \text{ kg/s}}$

3. (40 points)

1.5 kg of Air within a piston-cylinder assembly executes a Carnot heat pump cycle. For the cycle, $T_H = 600 \text{ K}$ and $T_C = 300 \text{ K}$. The energy rejected by heat transfer has a magnitude of 250 kJ per kg of air. The pressure at the start of the isothermal expansion is 325 kPa. Assuming the ideal gas model for the air, determine (a) the magnitude of the net work input, in kJ per kg of air, and (b) the pressure at the end of the isothermal expansion, in kPa. (5 pts)

- (5 pts) Draw a schematic of the system and label the system boundary.
- (3 pts) State the engineering model
- (5 pts) Write the simplified energy rate balance
- (10 pts) Determine the magnitude of the net work input, in kJ.
- (10 pts) Determine the pressure at the end of the isothermal expansion, in kPa.
- (7 pts) Sketch the process on a P-V diagram.



b) Engineering model:

- Carnot heat pump cycle $\cdot \Delta KE = \Delta PE = 0$
- Ideal gas
- Closed system

c) $\Delta E = Q - W$ $\Delta E = \Delta U + \overset{0}{\Delta KE} + \overset{0}{\Delta PE}$
 $\Delta E = \Delta U = Q - W$

d)

$$\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{300 \text{ K}}{600 \text{ K}} = 0.5$$

$$0.5 = \frac{Q_H}{\dot{W}} = \frac{250 \text{ kJ/kg}}{\dot{W}}$$

$$\dot{W} = 500 \text{ kJ/kg} \times 1.5 \text{ kg} = \boxed{750 \text{ kJ}}$$

e) $T_H = 600 \text{ K}$ $PV = RT$

$$V = \frac{RT}{P} = \frac{(8.314/28.97)(600 \text{ K})}{325 \text{ kPa}}$$

$$V = 0.5298 \text{ m}^3/\text{kg}$$

$$u @ 600 \text{ K} = 434.78 \text{ kJ/kg}$$

$$u @ 300 \text{ K} = 214.07 \text{ kJ/kg}$$

$$\Delta U = 220.71 \text{ kJ/kg} \quad \text{adiabatic}$$

$$-W = 220.71 \text{ kJ/kg} \quad \Delta U = -W$$

$$W = -220.71 \text{ kJ/kg} \quad W = \int P dV = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$220.71 \text{ kJ/kg} = (325 \text{ kPa})(V) + (325)(0.5298 \text{ m}^3/\text{kg})$$

$$V = 0.1493 \text{ m}^3/\text{kg}$$

$$P = \frac{(8.314/28.97)(600 \text{ K})}{0.1493 \text{ m}^3/\text{kg}} = \boxed{1153 \text{ kPa}}$$

